

Q1. Simplify $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$, with don't-care $d(w, x, y, z) = \Sigma(0, 2, 5)$, and show that an expression with the minimum number of literals is not necessarily unique.

EXAMPLE 3-12: Simplify the Boolean function:

$$F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$$

and the don't-care conditions:

While simplifying a Boolean function using a Karnaugh Map (K-map), the input combinations where the output is 1 are called minterms and are marked with 1s. Some inputs are not important—these are called don't-care conditions (X). These Xs are only used when they help form larger groups with the 1s. Larger groups mean fewer variables, which results in a simpler function. Finally, each group produces one product term, and all the terms together form the simplified Sum of Products expression.

$$F = w'z + yz$$

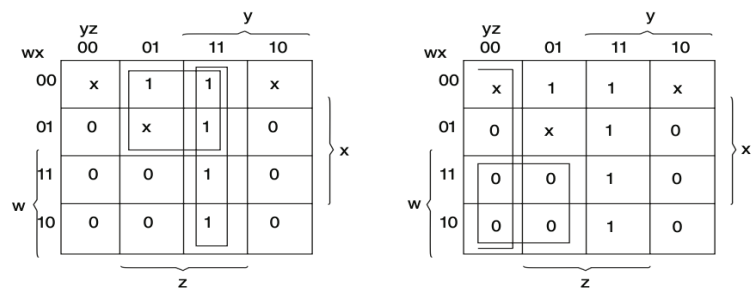
In (b), the 0's are combined with any X's convenient to simplify the complement of the function. The best results are obtained if we enclose the two X's as shown. The complement function is simplified to:

$$F' = z' + wy'$$

Complementing again, we obtain a simplified product of sums function:

$$F = z(w' + y)$$

The two expressions obtained in Example 3-12 give two functions which can be shown to be algebraically equal. This is not always the case when don't-care conditions are involved. As



(a) Combining 1's and X's $F = w'z + yz$ (b) Combining 0's and X's $F = z(w' + y)$

Figure 3.26 Example with don't-care conditions

a matter of fact, if an X is used as a 1 when combining the 1's and again as a 0 when combining the 0's, the two resulting functions will not yield algebraically equal answers. The selection of the don't-care condition as a 1 in the first case and as a 0 in the second results in different minterm expressions and thus different functions. This can be seen from Example 3-12. In the solution of this example, the X chosen to be a 1 was not chosen to be a 0. Now, if in Fig. 3-26(a) we choose the term $w'x'$ instead of $w'z$, we still obtain a minimized function:

$$F = w'x' + yz$$

But it is not algebraically equal to the one obtained in product of sums because the same X's are used as 1's in the first minimization and as 0's in the second.

This example also demonstrates that an expression with the minimum number of literals is not necessarily unique. Sometimes the designer is confronted with a choice between two terms with an equal number of literals, with either choice resulting in a minimized expression.

This example shows that the minimum number of letters in a minimized expression is not uniquely determined. Multiple solutions can be found based on the method of grouping or the use of don't-care conditions.

Q2: Simplify the Boolean function by using the **tabulation method**:
 $F = \Sigma(0, 1, 2, 8, 10, 11, 14, 15)$

Quine - McCluskey Method (Tabulation Method)

(a)

Group	w	x	y	z
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
8	1	0	0	0
10	1	0	1	0
11	1	0	1	1
14	1	1	1	0
15	1	1	1	1

(b)

Group	w	x	y	z
0, 1	0	0	0	-
0, 2	0	0	-	0
0, 8	-	0	0	0
2, 10	-	0	1	0
8, 10	1	0	-	0
10, 11	1	0	1	-
10, 14	1	-	1	0
11, 15	1	-	1	1
14, 15	1	1	1	-

Group	w	x	y	z
0, 2, 8, 10	-	0	-	0
0 8 2 10	-	0	-	0
10 11 14 15	1	-	1	-
10 14 11 15	1	-	1	-

$F = (wxy + xz) + (wy + wz)$
 prime implicant.

1st step: minterm ko binary me convert krj (0 → 0000, 5 → 0101 - - -)

1st group banate hai. karna? = 1st group ko 2nd group se compare krj (0 se 1 tak) 1 group, 2nd group, 3rd group, 4th group, 5th group, 6th group, 7th group, 8th group, 9th group, 10th group, 11th group, 12th group, 13th group, 14th group, 15th group.

step 2: 1st group ko 2nd group se compare krj (0 se 1 tak) 1 group, 2nd group, 3rd group, 4th group, 5th group, 6th group, 7th group, 8th group, 9th group, 10th group, 11th group, 12th group, 13th group, 14th group, 15th group.

step 3: 1st group ko 2nd group se compare krj (0 se 1 tak) 1 group, 2nd group, 3rd group, 4th group, 5th group, 6th group, 7th group, 8th group, 9th group, 10th group, 11th group, 12th group, 13th group, 14th group, 15th group.

4. Simplify the Boolean function $F(w, x, y, z) = \Sigma(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15)$

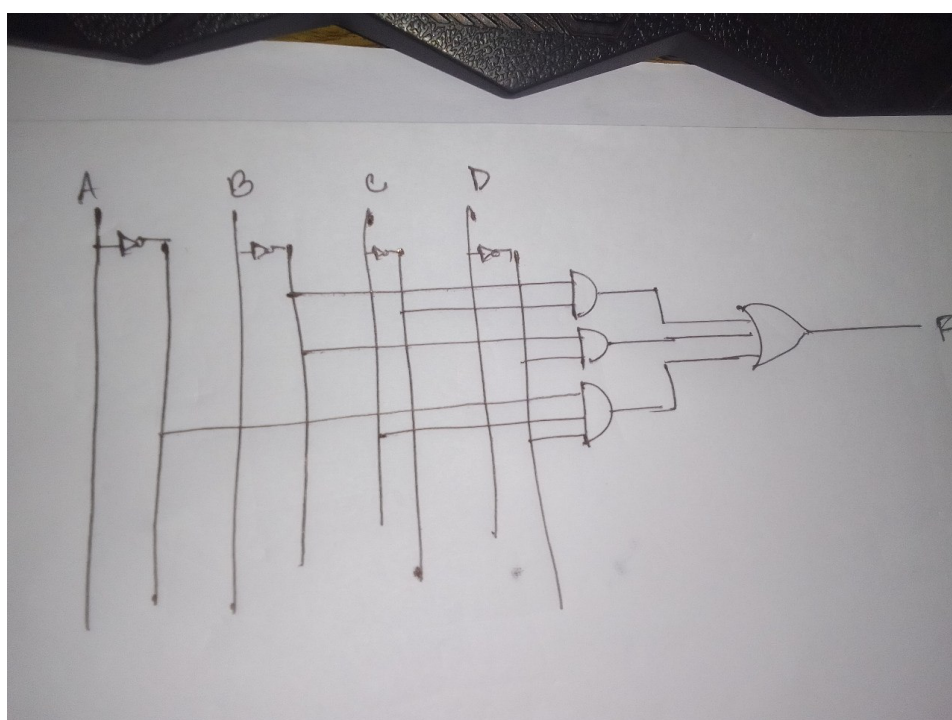
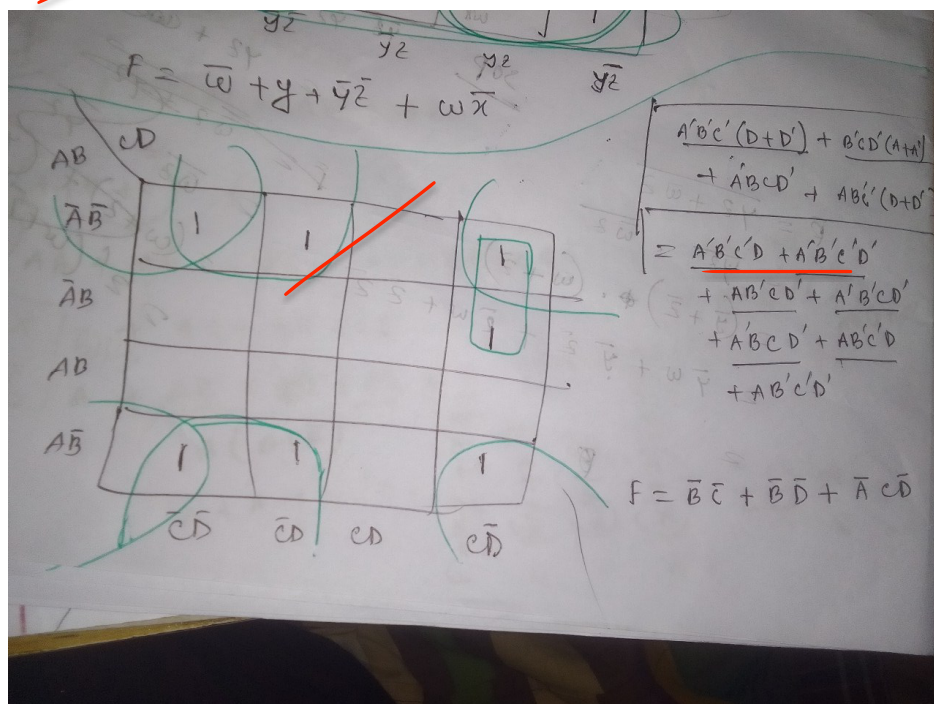
Answer:

$F(w, x, y, z) = \Sigma(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15)$

The Karnaugh map is a 4x4 grid with rows labeled $w\bar{x}$ and wx , and columns labeled $y\bar{z}$ and yz . The cells are filled with 1s for the minterms 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, and 15. The simplified expression is $F = \bar{w} + y + y\bar{z} + w\bar{x}$.

$F = \bar{w} + y + y\bar{z} + w\bar{x}$

5. Simplify $F = A'B'C'D + B'CD' + A'BCD' + AB'C'$ and implement the result using logic gates. ★



8. $F(A, B, C, D) = \Sigma(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$ — Show prime implicants and 7 essential prime implicants using Karnaugh map.

prime implicant : all possible group

Ex: $F(A, B, C, D) = \Sigma(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$

show prime implicant and 7 essential prime implicant.

64 Group

PI = $(A\bar{B}, CD, BD, AD, \bar{B}\bar{C}, \bar{B}\bar{D})$

EPI = $(\bar{B}\bar{C}, BD)$

Ans

NB: এক Group ১০ অর্ধে ছোট group হবে না।

আর $EPI \leq PI$
নাম বা কম হবে

9. Find the complement of the function $F_1 = xyz' + x'y'z'$ and $F_2 = x(yz' + yz)$ applying De Morgan's Theorem

$$F_1 = xyz' + x'y'z'$$

$$\overline{F_1} = \overline{(xyz' + x'y'z')}$$

Apply De Morgan's theorem

$$= \overline{xyz'} \cdot \overline{x'y'z'}$$

again apply.

$$= (\overline{x} + \overline{y} + z) \cdot (x + y + z)$$

Answer

$$F_2 = x(yz' + yz)$$

$$\overline{F_2} = \overline{x(yz' + yz)}$$

$$= \overline{x} + \overline{(yz' + yz)}$$

$$= \overline{x} + (\overline{yz'} \cdot \overline{yz})$$

$$= \overline{x} + (\overline{y} + z)(\overline{y} + \overline{z})$$

Answer.

10. What do you mean by complements? Make a comparison between 1's and 2's complements.

a **complement** is a way to represent the **negative** of a binary number using only 0s and 1s. It helps perform **subtraction** using **addition operations**.

There are two main types:

1. **1's Complement**
2. **2's Complement**

◆ 1's Complement:

👉 Flip all bits (0 becomes 1, 1 becomes 0)

Example:

$1010 \rightarrow 0101$

◆ 2's Complement:

👉 Flip all bits, then **add 1**

Example:

$1010 \rightarrow 0101 + 1 = 0110$

Feature	1's Complement	2's Complement
Rule	Flip bits	Flip bits + Add 1
Zero	Two (positive & negative)	One (only 0)
Subtraction	More complex	Simpler
Use in Computers	Rare	Common & preferred

12. Prove that $f_i = m_1 + m_2 + m_6 + m_7 = M_0 \cdot M_3 \cdot M_4 \cdot M_5$

$$f_i = m_1 + m_2 + m_6 + m_7$$

$$F(ABC) = m_1 + m_2 + m_6 + m_7$$

↓
Karnaugh Map

$$\overline{F(ABC)} = m_0 + m_3 + m_5 + m_7$$

↓
000 011 100 101

again
complement

$$\begin{aligned} \overline{\overline{F(ABC)}} &= \overline{\overline{ABC}} + \overline{\overline{AB\overline{C}}} + \overline{\overline{A\overline{B}C}} + \overline{\overline{A\overline{B}\overline{C}}} \\ &= \overline{\overline{ABC}} \cdot \overline{\overline{AB\overline{C}}} \cdot \overline{\overline{A\overline{B}C}} \cdot \overline{\overline{A\overline{B}\overline{C}}} \\ &= (A+B+C) \cdot (A+B+\overline{C}) \cdot (\overline{A}+B+C) \cdot (\overline{A}+B+\overline{C}) \end{aligned}$$

0 0 0 0 1 1 1 0 0 1 0 0

$$F(ABC) = m_0 \cdot m_3 \cdot m_4 \cdot m_5$$

So $m_1 + m_2 + m_6 + m_7 = m_0 \cdot m_3 \cdot m_4 \cdot m_5$ (proved)

14. Express the following function in product of maxterm form: $f(w, x, y, z) = y'z + wxz' + w'x'z + w'x'z$ short

$$F(w, x, y, z) = \bar{y}z + wxz' + w'x'z$$

$$\bar{y}z(w + w')$$

$$= w\bar{y}z + w'\bar{y}z$$

$$= w\bar{y}z(x + \bar{x}) + w'\bar{y}z(x + \bar{x})$$

$$= \underline{wx\bar{y}z} + \underline{w\bar{x}\bar{y}z} + \underline{w'x\bar{y}z} + \underline{w'\bar{x}\bar{y}z}$$

product of MAX

$$\begin{aligned} wxz'(y + \bar{y}) \\ = \underline{wxzy\bar{z}} + \underline{wxz\bar{y}\bar{z}} \end{aligned}$$

$$\begin{aligned} \bar{w}\bar{x}z(y + \bar{y}) \\ = \underline{\bar{w}\bar{x}zy\bar{z}} + \underline{\bar{w}\bar{x}z\bar{y}\bar{z}} \end{aligned}$$

SOP = $\Sigma(1, 3, 5, 9, 12, 13, 14)$
 POS = $\Pi(0, 2, 4, 6, 7, 8, 10, 11, 15)$

Answer

	$y\bar{z}$	$\bar{y}z$	$y\bar{z}$	$\bar{y}\bar{z}$
$w\bar{x}$		1	1	
$w\bar{x}$		1		
wx	1	1		1
$w\bar{x}$		1		

$$F = wx\bar{z} + \bar{w}\bar{x}z + \bar{y}z$$

$$\begin{aligned} F &= \overline{wx\bar{z}} + \overline{w\bar{x}z} + \overline{\bar{y}z} \\ &= \overline{wx\bar{z}} \cdot \overline{w\bar{x}z} \cdot \overline{\bar{y}z} \\ &= (\underline{w + \bar{x} + z}) (\underline{\bar{w} + x + \bar{z}}) (\underline{y + \bar{z}}) \end{aligned}$$

16. What are the **advantages and disadvantages** of digital techniques over analog techniques?

✓ **Advantages of Digital:**

1. Less noise – clear signal
2. Easy to store and save
3. Accurate and reliable
4. Easy to copy without losing quality
5. Good for computers
6. More secure (can be encrypted)

✗ **Disadvantages of Digital:**

1. Needs more space (bandwidth)
2. Circuits are more complex
3. Some detail lost when converting from analog
4. May use more power
5. Setup can be costly
6. Can be slower sometimes

17. What is **duality principle**? Prove $x(x+y) = x$ by duality.

The **duality principle** states that every Boolean expression has a dual expression. The dual of a Boolean expression can be obtained by:

- Replacing AND ($\cdot$) with OR ($+$)
- Replacing OR ($+$) with AND ($\cdot$)
- Replacing 1 with 0 and 0 with 1.

To prove: $x(x+y)=x$ duality

$$=x+(x.y) \text{ [we know that } x+(x.y)=x]$$

$$=x \text{ (proved)}$$

18. Represent the decimal number 8620:

- (i) in **biquinary**
- (ii) in **excess-3 code**
- (iii) in **2-4-2-1 code**

(i) Biquinary Representation:

Biquinary রিপ্রেজেন্টেশন হল প্রতিটি ডিজিটকে **দুইটি বাইনারি সংখ্যা** দ্বারা উপস্থাপন করা। একটি ১০-এর স্থান এবং একটি ১-এর স্থান।

প্রক্রিয়া:

- 8 → 1000 (10-এর স্থান) এবং 0001 (1-এর স্থান)
- 6 → 0110 (10-এর স্থান) এবং 0000 (1-এর স্থান)
- 2 → 0010 (10-এর স্থান) এবং 0000 (1-এর স্থান)
- 0 → 0000 (10-এর স্থান) এবং 0000 (1-এর স্থান)

ফলস্বরূপ:

Biquinary=1000 0001 0110 0000

(ii) Excess-3 Code:

Excess-3 কোডে, প্রতিটি ডিজিটে **৩ যোগ** করা হয় এবং তারপরে বাইনারিতে রূপান্তর করা হয়।

প্রক্রিয়া:

- 8 → $8+3=11$ → 1011 (বাইনারি)
- 6 → $6+3=9$ → 1001 (বাইনারি)
- 2 → $2+3=5$ → 0101 (বাইনারি)
- 0 → $0+3=3$ → 0011 (বাইনারি)

ফলস্বরূপ:

Excess-3=1011 1001 0101 0011

(iii) 2-4-2-1 Code:

2-4-2-1 কোডে, প্রতিটি ডিজিটকে **৪টি বিট** দ্বারা উপস্থাপন করা হয়, যেখানে বিটগুলোর ওজন যথাক্রমে ২, ৪, ২, এবং ১।

প্রক্রিয়া:

- $8 \rightarrow 10001\ 0\ 0\ 0\ (2 + 4 + 2 + 1)$
- $6 \rightarrow 01100\ 1\ 1\ 0\ (4 + 2)$
- $2 \rightarrow 00100\ 0\ 1\ 0\ (2)$
- $0 \rightarrow 00000\ 0\ 0\ 0$

ফলস্বরূপ:

$$2-4-2-1=1000\ 0110\ 0010\ 0000$$

19. Convert octal **623.77** to:

- decimal
- binary
- hexadecimal

20. Convert $(511)_4$ to base-5.

Summary of Results:

1. **623.77₈ in Decimal:** 403.984375
2. **623.77₈ in Binary:** 110010011.111111.
- 623.77₈ in Hexadecimal:** C93.F₁₆

20 Convert $(511)_4$ to base-5:

Step 1: Base-4 to Decimal

$$511_4 = (5 \times 4^2) + (1 \times 4^1) + (1 \times 4^0) = 80 + 4 + 1 = 85$$

Step 2: Decimal to Base-5

$$\begin{array}{ll} 85 \div 5 = 17 & \rightarrow \text{remainder} = 0 \\ 17 \div 5 = 3 & \rightarrow \text{remainder} = 2 \\ 3 \div 5 = 0 & \rightarrow \text{remainder} = 3 \end{array}$$

Answer:

$$(511)_4 = (320)_5$$

23. Simplify the Boolean function F using the don't-care conditions:

$$F = w'(x'y + x'y' + xyz) + x'z'(y + w)$$

$$d = w'x(y'z + yz') + wyz$$

Handwritten solution for simplifying the Boolean function F using don't-care conditions.

Given function: $F = w'(x'y + x'y' + xyz) + x'z'(y + w)$

Don't-care conditions: $d = w'x(y'z + yz') + wyz$

Step 1: Expand F and d into sum of minterms.

For F:

$$F = w'(x'y + x'y' + xyz) + x'z'(y + w)$$

$$F = w'x'y + w'x'y' + wxyz + x'y'z' + wx'z'$$

For d:

$$d = w'x(y'z + yz') + wyz$$

$$d = w'xy'z + w'xyz' + wyz$$

Step 2: Karnaugh Map (K-map) for F and d.

Variables: w, x, y, z. The K-map is a 4x4 grid with rows labeled w (0, 1, 2, 3) and columns labeled x (0, 1, 2, 3).

Map for F (minterms 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15):

0	1	1	1
0	1	1	1
0	1	1	1
0	1	1	1

Map for d (minterms 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15):

0	1	1	1
0	1	1	1
0	1	1	1
0	1	1	1

Step 3: Simplify F and d using K-map.

For F:

- Group 1: $w'x'y$ (minterms 1, 2, 3, 4)
- Group 2: $w'x'y'$ (minterms 5, 6, 7, 8)
- Group 3: $wxyz$ (minterms 9, 10, 11, 12)
- Group 4: $x'y'z'$ (minterms 13, 14, 15, 0)
- Group 5: $wx'z'$ (minterms 1, 2, 3, 4)

For d:

- Group 1: $w'xy'z$ (minterms 1, 2, 3, 4)
- Group 2: $w'xyz'$ (minterms 5, 6, 7, 8)
- Group 3: wyz (minterms 9, 10, 11, 12)

Step 4: Final simplified expression.

For F:

$$F = w'x'y + w'x'y' + wxyz + x'y'z' + wx'z'$$

For d:

$$d = w'x(y'z + yz') + wyz$$

Final Answer: $F = w'x'y + w'x'y' + wxyz + x'y'z' + wx'z'$